



How Black Holes shape galaxies

New data has revealed how supermassive black holes shape galaxies with strong winds. <https://digit.in/aug19-01>



LightSail 2 has been deployed

The spacecraft by Planetary Society will harness the energy of the sun to change its orbit around Earth. <https://digit.in/aug19-02>

A simulation of a what a miniature black hole decay event would look like within the ATLAS experiment at the LHC

HYPOTHETICAL PARTICLES

Credit: CERN

We have discovered the Higgs Boson, what is next?

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he discovery of the Higgs Boson in 2012 was the final piece of the puzzle in the Standard Model, the

theory that is widely accepted to explain almost everything we understand about the physical universe. The Standard Model predicted the existence of such a particle, but

without confirmation of its existence, researchers had to come up with an explanation for why the particle did not exist. When announcing the discovery, CERN physicists made it clear that the discovered particle could only be the lightest of a family of Higgs Bosons, and that there might be other particles to be found, that cannot be fully explained by the Standard Model. After discovering the expected particles in the Standard Model, researchers are now looking at physics beyond the Standard Model. High energy particle physicists are now in the process of hunting for exotic particles that do not fit in the Standard Model, perhaps paving the way for a more fundamental understanding of nature. These searches are guided by

simulations based on models other than the Standard Model, which give an idea to the scientists on what kind of interactions to expect within the particle accelerators. As the search involves combing through mountains (or petabytes) of data on various collisions events, there are already some candidates identified for some of the exotic particles hypothesised to exist. Within these events, researchers may even find particles that no current theories or models have anticipated. These experiments may one day help answer questions that science cannot at this point of time, such as: Why there is more matter than antimatter in the universe? What makes up dark matter? Why does the Higgs Boson weigh as much as it does? Are there more than